

where β is the coefficient from step 1, and $\varepsilon_{\text{base}}$ is the efficiency in the first historical year.

4. **Replace the useful exergy-GDP link** in cells `BAU!I71:AJ71` and `Decarbonizatio!I73:AJ73`:

- If intensity is constant: $\text{GDP} * \text{mean_intensity}$.

- If not constant: use the extrapolation formula from the Eff-TFP kit.

5. **Update historical and projected input data** (renewable capacities, demographics, sectoral exergy shares, efficiencies) to match your country.

6. **All dependent cells recalculate automatically** – the model is now ready for scenario analysis.

Comments & Notes (as found in the Excel file)

The Excel file contains inline comments that explain the purpose of each sheet and column:

- **`1\_Standard\_Template`** (if present) – “Standard Mode — provide all macro + exergy columns. The kit computes TFP from scratch using factor shares, labour, capital, and GDP. Yellow = optional (used if available; analysis proceeds without them).”

- **`2\_Preloaded\_Template`** – “Pre-loaded Mode — use this sheet if TFP and efficiency series are already available ... Series will be normalized to $z = \ln(x / x_0) + 1$ if not already done.”

- In the `BAU` sheet, the top rows include: **“Scenarios - BAU 2025-2050”** and column headers such as **“New Solar power [GW]”**, **“Cost per GW”**, **“Investment Industry [Mrd Euros]”**, **“Labour [Hours worked by engaged people]”**, etc.

- In the `Decarbonizatio` sheet: **“Scenarios up to 2050 - National Roadmap 2050”** and **“Extra ... compared to BAU”** rows show the additional deployment and cost relative to the baseline.

Dependencies

- **Microsoft Excel** (any version that supports standard formulas, no VBA required).

- No external data files – all input data are embedded in the workbook.

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Contact

For questions or to report issues, please contact the MAPS project team via the project website or the corresponding author of D4.2.